

Math/Stat 452: Mathematical Statistics

Instructor: Benjamin Ridenhour

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Office Hours: By appointment

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Class Hours: MWF 8:30–9:20am

Class Room: JEB 026

Course Description

In this course, we will focus on understanding the interplay between probability, statistics, and modeling. In many regards, probability and statistics are complementary branches of applied mathematics. Probability expresses the chance of observing an outcome given a model (e.g., a distribution); conversely, statistics asks whether a given set of observations are consistent with a parameterization of a chosen mathematical model given an assumed probability distribution. More specifically, in statistics we often seek to find a range of parameter values in which we are confident or to distinguish between competing models with some degree of certainty. Likewise, “inverse problems” in mathematics attempt to find valid parameters given a generative model and observed data, which is simply another way to discuss the practice of statistical model fitting. This course will help you understand and implement the tools required to build complex models and fit them to real-world data.

Course Structure

Our class is intended to be an in-person class, and, as such, content delivery will be done via a live lecture. Obviously, students will not always be able to attend class; I will do my best to accommodate situations as they arise. I recommend you attend if at all possible. Similarly, evaluation (exams, quizzes) are being done in-person (see below). This does not apply to engineering outreach (EO) students.

Coursework will be as follows. Chapters of the textbook are fairly short, so we will typically only spend once class period per chapter. Suggested homework problems will be given for each section of the book; problem sets will be posted to Canvas. Answers to the suggested problems can be found in the back of the book. Friday class periods will partially be used for quizzes. **The graded portion of the class will be your performance on three equally-weighted exams, in addition to the Friday quizzes.** In other words, each evaluated facet of class is worth 25% of your overall grade. I expect to give 9 quizzes of which you can drop one (so 8 quizzes count toward your grade). Quizzes will cumulatively count as one exam. You will be able to drop one

of the four exams; only the final exam is cumulative. Exams will roughly cover about 9 chapters (except for the fourth exam). *You will be allowed to have a page of notes during each exam, but no calculators are allowed!* **If you cannot be in attendance for a quiz or exam, arrangements need to be made with me prior to the event.**

Required Materials

- *Probability and Mathematical Statistics*. By MC Meyer. Society for Industrial and Applied Mathematics: Philadelphia.

Prerequisites/Corequisites

Prerequisites: Math/Stat 451

Topics By Week (Approximate)

Note that session numbers (e.g., session 1 is noted §1) are given for EO students and are also approximate. *As mentioned above, quizzes are given at the beginning of class of Fridays.*

1. Wks 1–2; §1–4: Introduction and review of distributions (discrete, continuous, multivariate); *No class MLK day 15 Jan.*
2. Wk3; §5: R, §6: Ch. 42 (Parameter Estimation), §7: Ch. 43 (MLE)
3. Wk 4; §8: Ch. 43 (MLE cont.), §9 & §10: Ch. 44 (MM; W&F)
4. Wk 5; §11: Review, **Test 1**, §12: Ch. 46 (Consistency)
5. Wk 6; §13: Ch. 50 (CIs), §14: Ch. 51 (Bootstrapping), §15: Ch. 52 (Information)
6. Wk 7; §16: Ch. 53 (Sufficient Statistics), §17: Ch. 54 (UMVUE); *No class President's day, 19 Feb.*
7. Wk 8; §18: Ch. 55 (Exponential Family), §19: Numerical methods, §20: Review
8. Wk 9; **Test 2**, §21 Test 2 Summary, *No class 10 Mar.*
9. Wk 10; *Spring Break 11–15 Mar.*
10. Wk 11; §22: Ch. 34 (Sampling a Normal), §23: Ch. 35 (Hypothesis Testing), §24: Ch. 36 (Hypothesis Testing)
11. Wk 12; §25: Ch. 56 (Size and Power), §26: Ch. 57 (Neyman-Pearson Lemma), §27: Ch. 58 (LRT)
12. Wk 13; §28: Ch. 58 (LRT cont.), §29: CIs and testing, §30: Review
13. Wk 14; **Test 3**, §31 & §32: Ch. 59 (χ^2 -tests; W&F)

14. Wk 15; §33 & §34: Models relying on normal distributions (M&W), §35: Ch. 60 (ANOVA)
15. Wk 16; §36: Ch. 45 (Bayesian Estimation), §37 & §38: Ch. 49 (Credible Intervals; W&F)
16. Wk 17; §39: Model selection, Hierarchical models, §40: MCMC, §41: Review

Course Objectives/Learning Outcomes

Successful students will:

1. Understand the theory of statistics sufficiently to solve most problems appearing in standard textbooks for this course.

Final Exam Date

10 May (Friday), 8:00–10:00am

Grading Policy

A typical grading scale will be used with cutoffs every 10% (i.e., $\geq 90\% = A$, $(90\%, 80\%] = B$, ...). I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments using the following proportions:

- 25% of your grade will be determined by 3 of the 4 exams (lowest exam by percentage is dropped).
- 25% of your grade will be determined by 8 of your 9 quiz scores (lowest quiz by percentage is dropped).

Engineering Outreach Due Dates

All quizzes and exams will be due one week after their actual delivery in class. For example, if a test was given on a Wednesday, EO students will have until the next Wednesday to have a proctored exam taken and returned for grading.